MONTH 1 — PYTHON FOUNDATIONS

DAY 1: Introduction to Python — Variables, Data Types & print()

Week 1 · Python Basics · Beginner → Practical AI Engineer Track

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1. Theory

Python is a high-level, interpreted programming language used across AI, data science, web development, and automation. It's the primary language used throughout this entire course because virtually every major AI/ML library (NumPy, Pandas, PyTorch, TensorFlow, Hugging Face) is built for it.

1.1 Why Python for AI Engineering

- Simple, readable syntax — close to plain English.
- Massive ecosystem of AI/ML libraries.
- Huge community support and documentation.
- Works for scripting, backend, data pipelines, and model training.

1.2 Variables

A variable is a named container that stores a value in memory. Python is dynamically typed, meaning you don't declare a type — Python infers it automatically from the value you assign.

- Variable names are case-sensitive (age and Age are different).
- Must start with a letter or underscore, not a digit.
- Cannot use Python reserved keywords (e.g. class, for, if).
- Convention: use snake_case for variable names (student_name, not StudentName).

1.3 Core Data Types

- int — whole numbers, e.g. 25, -3, 1000
- float — decimal numbers, e.g. 3.14, -0.5
- str — text, written in quotes, e.g. "AI Engineer"
- bool — True or False
- NoneType — represents "no value", written as None

1.4 print() and Comments

print() displays output to the console — it's the most-used function for checking what your code is doing. A comment (starting with #) is text Python ignores; it's used to explain code to humans reading it.

2. Syntax

Declaring variables and printing values:

```
# Variable assignment — no type keyword needed
name = "Ayesha"        # str
age = 22               # int
height = 5.6           # float
is_student = True      # bool
mentor = None          # NoneType

# print() displays values
print(name)
print("Age:", age)

# type() reveals the data type
print(type(age))       # <class 'int'>

# f-strings: the modern way to build text with variables
print(f"{name} is {age} years old.")

# input() reads text typed by the user (always returns a string)
city = input("Enter your city: ")
print("You live in", city)
```

3. Small Example

A short program that introduces itself using variables and an f-string:

```
course = "AI Engineering"
day = 1
hours_today = 2

print(f"Welcome to Day {day} of the {course} course!")
print(f"Today's focused practice time: {hours_today} hours.")
print("Goal: get comfortable with variables, data types, and print().")
```

Expected output:

```
Welcome to Day 1 of the AI Engineering course!
Today's focused practice time: 2 hours.
Goal: get comfortable with variables, data types, and print().
```

4. 2-Hour VS Code Practical

Open VS Code, create a folder called day01, and inside it create two files: program1.py and program2.py. Run each with the ▶️ Run button or python program1.py in the terminal. Aim for roughly 1 hour per program — read the task, write the code yourself before checking the Solutions section.

Program 1 (≈60 min) — Personal Profile Card

Write a program that:

1. Stores your name, age, city, and favorite programming language in four variables.
2. Stores a boolean variable is_learning_ai set to True.
3. Prints all details in a neatly formatted "profile card" using f-strings, e.g.:

```
===== PROFILE CARD =====
Name        : Ayesha
Age         : 22
City        : Lahore
Language    : Python
Learning AI : True
========================
```

Bonus: use type() to print the data type of each variable underneath the card.

Program 2 (≈60 min) — Simple Interactive Calculator Intro

Write a program that:

4. Uses input() to ask the user for their name and age (age will come in as a string).
5. Converts age to an int using int().
6. Calculates and prints what year they were roughly born in (use 2026 as the current year).
7. Prints a friendly summary sentence using an f-string.

Example run:

```
Enter your name: Bilal
Enter your age: 24
Hi Bilal! You were likely born around the year 2002.
```

5. Solutions

Solution — Program 1

```
name = "Ayesha"
age = 22
city = "Lahore"
language = "Python"
is_learning_ai = True

print("===== PROFILE CARD =====")
print(f"Name        : {name}")
print(f"Age         : {age}")
print(f"City        : {city}")
print(f"Language    : {language}")
print(f"Learning AI : {is_learning_ai}")
print("========================")
```

```
# Bonus
print(type(name), type(age), type(city), type(language),
type(is_learning_ai))
```

Solution — Program 2

```
name = input("Enter your name: ")
age_str = input("Enter your age: ")
age = int(age_str)             # convert text -> whole number

current_year = 2026
birth_year = current_year - age

print(f"Hi {name}! You were likely born around the year {birth_year}.")
```

Common beginner mistake: forgetting int(age_str) and trying current_year - age_str directly — Python cannot subtract a string from a number, so it raises a TypeError.

6. Mini Project — "My AI Journey" Intro Script

Combine everything from today into one small script, ai_journey.py:

8. Ask the user for their name and current profession using input().
9. Store today's date info as variables: course_day = 1, total_days = 168.
10. Calculate progress percentage: (course_day / total_days) * 100, rounded to 2 decimal places using round().
11. Print a motivational summary using an f-string, e.g. "Hi <name>! You're a <profession> starting Day 1 of 168 (0.6% complete). Let's build toward AI Engineer level!"

This mini project reinforces variables, data types, type conversion, input/output, and f-strings — the exact building blocks every later day (Week 1–24) will build on.

Next: Day 2 will build on this with arithmetic operators, string methods, and basic user-input validation.